

Supplementary Information

Target-conditioned refinement of future-blind longitudinal glioma segmentation-change maps across independent cohorts

[AUTHOR_LIST_REQUIRED]

Supplementary Methods

Development P0 used patient-disjoint out-of-fold inference. Independent internal and RHUH P0 maps used equal-weight ensembles of five frozen checkpoints. EIA and target-volume-matched top-k analyses are retrospective target-access controls. Full patient linkage, authority manifests, code hashes and audit logs are supplied in the accompanying reproducibility archive.

Supplementary Results

Tables S1–S10 contain concise scientific summaries required to evaluate the manuscript. They do not introduce a new endpoint or inferential comparison. Repository-scale engineering inventories and full SHA-256 manifests were intentionally moved to the reproducibility archive.

Supplementary Table S1. Cohort metadata availability and analysis flow

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

Characteristic
Source patients / pre-outcome excluded / analysed
Age, years, median (IQR)
Sex, female / male
Current-to-future interval, days, median (IQR)
Diagnosis / grade
IDH status
Extent of resection
Previous treatment

Supplementary Table S2. Independent internal secondary metric summaries

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

method	metric	n	mean	SD	median	Q1	Q3	bootstrap_95ci_low	bootstrap_95ci_high
Fixed	IoU_0.5	113	0.14275676271633445	0.09133695791297426	0.1362596937760986	0.0802512829587683	0.1871854901644196	0.12651103108703468	0.16008221968616135
Fixed	soft_Dice	113	0.18515205004391988	0.11200243718867303	0.1836706572019499	0.1030116628668069	0.2431405016098808	0.16498177393457839	0.2062280610094493
Fixed	Brier	113	0.004523438996233918	0.004734199430969551	0.0033030287317839	0.0016272471458048	0.0051209919813546	0.0037188075733762386	0.005440357760197517
Fixed	average_precision	113	0.22720634123309832	0.16241971610825343	0.1879753874478399	0.0985860560241737	0.3365579559499516	0.19772194861587575	0.2580833335240748
Full PCC	IoU_0.5	113	0.3026637349790633	0.1497703708873603	0.3067127425104078	0.2104066683505935	0.3918147956056391	0.2752864707117545	0.33016035876785954
Full PCC	soft_Dice	113	0.26671044302084707	0.14958672757899014	0.2634951748915842	0.1626158578635101	0.3671391018662227	0.23960427941804643	0.29425261655770213
Full PCC	Brier	113	0.0031018553812365104	0.004117262708921748	0.001720032669824	0.0008421584371434	0.0032308057301018	0.002413682840374779	0.00392091604782403
Full PCC	average_precision	113	0.41423683094124614	0.21167656087785178	0.4165520070514784	0.2551603624748501	0.5815393189258564	0.3756837871707996	0.4536073911976625
No-smoothing PCC	IoU_0.5	113	0.49576499308548405	0.2014320336851955	0.5164104694640631	0.3629558810869102	0.6289949050486336	0.458882042538722	0.532953099563842
No-smoothing PCC	soft_Dice	113	0.32036574405808416	0.16473475891142472	0.3258734814635898	0.2143762502949238	0.4415489769209107	0.29049068785040427	0.35058833388603805
No-smoothing PCC	Brier	113	0.0025827483372998622	0.004011020143365503	0.0012057733558566	0.0005939081473493	0.002498761252655	0.001913045595018462	0.0033877414396320553
No-smoothing PCC	average_precision	113	0.6473279573040465	0.19877701117705623	0.6639420594536367	0.5325590073365468	0.8112320862225844	0.6107317542528754	0.6825288889115569

Supplementary Table S3. RHUH secondary metric summaries

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

method	metric	n	mean	SD	median	Q1	Q3	bootstrap_95ci_low	bootstrap_95ci_high
Fixed	Dice_0.5	39	0.18950017890741766	0.13181373260495102	0.1810851082478745	0.08101134869194154	0.2697179531632183	0.14895061636799545	0.23064898178709717
Fixed	IoU_0.5	39	0.11054095941779987	0.08301115651137661	0.09955666923670008	0.04222215768252242 4	0.1558935171991167	0.08506311538024823	0.13648685381671424
Fixed	soft_Dice	39	0.1511701565526571	0.10643483811300757	0.14559822823352356	0.06034975409133125 4	0.22688237485507257	0.11815825590103143	0.18381377226935386
Fixed	Brier	39	0.00928385394496658	0.00469905463170288 7	0.00763471577763848	0.00545406254663618 1	0.01239327355435607 1	0.00785386751634234 4	0.01076272645874931 5
Fixed	average_precision	39	0.1530624640609824	0.1213144906077636	0.14168728737955094	0.04085305084152732	0.23865676937197935	0.11615144634123438	0.1910199847596164
Full PCC	Dice_0.5	39	0.36542690071149336	0.1877101209161402	0.39133898182344845	0.22914512402212117	0.5174705877862634	0.30641114953412724	0.42139239015738184
Full PCC	IoU_0.5	39	0.23890190219900248	0.13790871235348484	0.24327000990366435	0.12946013210166016	0.3490646684277705	0.19586371363715865	0.280464200337404
Full PCC	soft_Dice	39	0.23822373120425466	0.14449822461110426	0.2284472165006823	0.13395987203032744	0.3584749960855774	0.19351501347826133	0.2817220311880662
Full PCC	Brier	39	0.00645728681519453 1	0.00413205628951052 4	0.00506184843031575 4	0.00308286583574477 9	0.00827878149673554 7	0.00522995159182881 6	0.00777512379358717 35
Full PCC	average_precision	39	0.309313693664386	0.1903034387517972	0.3213075313687235	0.15034673313328795	0.489838043981093	0.25021657937664316	0.3676327473641208
No-smoothing PCC	Dice_0.5	39	0.45122727714465866	0.2103476967944721	0.4657538819314243	0.35019353155385713	0.6201061384084978	0.38532966869418456	0.5142037632187004
No-smoothing PCC	IoU_0.5	39	0.31359068073945195	0.16979067258624664	0.3035718170939551	0.21254294979186955	0.44938745365788635	0.2613944366050698	0.36525906695264837
No-smoothing PCC	soft_Dice	39	0.269638303937967	0.15574532496540452	0.27767959662906516	0.16528694729409016	0.4002673614695276	0.22149456684742033	0.31654168401660904
No-smoothing PCC	Brier	39	0.00576466158045133 1	0.00395946527078449 8	0.00468596687477208 4	0.00267474815627309 9	0.00750315880314495 8	0.00459694865485668 9	0.00704479139581632
No-smoothing PCC	average_precision	39	0.40436480206898673	0.20907832050513858	0.4197155058804299	0.2724584123521825	0.5755222215433748	0.33964211645401166	0.4687456176001924

Supplementary Table S4. Development fixed-threshold and probability metric summaries

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

method	metric	mean	SD	median	Q1	Q3	bootstrap_95ci_low	bootstrap_95ci_high
FIXED	dice_fixed	0.21556589877238333	0.16057018126125533	0.17748534359324275	0.09120109104428663	0.2974687837566823	0.16985336051023686	0.2674448003235076
FIXED	soft_dice	0.22223404839292007	0.147081354913252	0.2052862059284397	0.09413591471964063	0.28970313001623493	0.17934569141838044	0.26881379052376175
FIXED	brier_score	0.006301912736929374	0.0028138783705253362	0.00580276136415055	0.004003951752135875	0.008300134343036199	0.005451139892318402	0.007166331826087403
FIXED	average_precision	0.2293721052409144	0.18719922765289493	0.1735853639516701	0.09479573938614945	0.35398556847904783	0.17588411988712874	0.28979215319490303
FULL_PCC	dice_fixed	0.32575423583519675	0.1871118096506793	0.2672570518157462	0.18834634832976513	0.43879100545898797	0.2706213325910587	0.3857240353811594
FULL_PCC	soft_dice	0.26986480442716926	0.16020259264743347	0.2583983758673641	0.1380498078249373	0.35398810419650434	0.22190333430321788	0.320206524882015
FULL_PCC	brier_score	0.004702401769835172	0.0022371192227205613	0.0044005255538703	0.00297187375242395	0.006230816184005775	0.004027149191587236	0.005403257533925778
FULL_PCC	average_precision	0.3730950537963239	0.20958885063734023	0.36369438870739224	0.19648824527225672	0.5115405353256944	0.31172187357695474	0.4392034975285071
NO_SMOOTHING	dice_fixed	0.3850225818756014	0.21473830504516306	0.35757261204125057	0.20516524279679194	0.5019559011481005	0.32098254232830964	0.4527167131310237
NO_SMOOTHING	soft_dice	0.30945156070623653	0.1716954111405503	0.30212541676869825	0.18224531960304988	0.3939770182368503	0.25849273769128017	0.3633043190954102
NO_SMOOTHING	brier_score	0.0038531325669077	0.0019602598924239765	0.0034951734711370502	0.0023487904712739	0.004817403451139325	0.0032805126188293126	0.004463492086587708
NO_SMOOTHING	average_precision	0.5231435278147645	0.20241743002203985	0.518723843461515	0.382499142537183	0.6517454172608723	0.4616220881476936	0.585647450760667

Supplementary Table S5. Development mechanism ablations under oracle-assisted target-volume-matched top-k Dice

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

condition	method	metric	N	mean	SD	median	Q1	Q3	bootstrap_95ci_low	bootstrap_95ci_high
FACTORIAL_ERROR_OFF_SUPPRESSION_OFF		dice	40	0.2762071661327381	0.16648155812622573	0.26824731584193473	0.1605030356699344	0.39412070110037045	0.22646387018022207	0.3291898900404981
FACTORIAL_ERROR_OFF_SUPPRESSION_ON		dice	40	0.28807542785274454	0.16609163800542953	0.2717145150435562	0.16296507290500262	0.41257797576110855	0.23917261534297019	0.3402789807346136
FACTORIAL_ERROR_ON_SUPPRESSION_OFF		dice	40	0.36124881606469395	0.1783961471757867	0.3648097254845589	0.22398566567481387	0.4674398796645586	0.30873856225771446	0.41657250007476243
FIXED_P0		dice	40	0.2762071661327381	0.16648155812622573	0.26824731584193473	0.1605030356699344	0.39412070110037045	0.22718335196982162	0.3283168501329602
FULL_PCC		dice	40	0.3884208514831534	0.1735083320190867	0.37746157603644065	0.2644963101871098	0.49369194950449413	0.336739031093995	0.44278639014116333
GLOBAL_DISCREPANCY		dice	40	0.39322673515301176	0.1714978248066687	0.3786466054545023	0.2786469540144292	0.49829997289273187	0.3410055061355568	0.44774499786358746
NO_SMOOTHING		dice	40	0.5001397076361601	0.17635167273893507	0.5133707035864199	0.378782574342467	0.5945198148192297	0.4466413800239759	0.5544621884725907

Supplementary Table S6. Imperfect-guidance robustness

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

condition	method	n	oracle_assisted_topk_Dice_mean
CLEAN	PCC	40	0.388421
CLEAN	EIA_LINEAR	40	0.349947
CLEAN	EIA_BLEND_075	40	0.532841
PARTIAL_50	PCC	40	0.349182
PARTIAL_50	EIA_LINEAR	40	0.339260
PARTIAL_50	EIA_BLEND_075	40	0.492284
PARTIAL_25	PCC	40	0.323722
PARTIAL_25	EIA_LINEAR	40	0.327635
PARTIAL_25	EIA_BLEND_075	40	0.459638
FP_25	PCC	40	0.384170
FP_25	EIA_LINEAR	40	0.343099
FP_25	EIA_BLEND_075	40	0.532171
SHIFT_3	PCC	40	0.368439
SHIFT_3	EIA_LINEAR	40	0.340536
SHIFT_3	EIA_BLEND_075	40	0.484517
MIXED	PCC	40	0.332755
MIXED	EIA_LINEAR	40	0.324969
MIXED	EIA_BLEND_075	40	0.454213

Supplementary Table S7. No-smoothing robustness across guidance conditions

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

condition	method	metric	N	mean	SD	median	Q1	Q3	bootstrap_95ci_low	bootstrap_95ci_high
CLEAN	FULL_PCC	dice_topk	40	0.3884208514831534	0.1735083320190867	0.37746157603644065	0.2644963101871098	0.49369194950449413	0.3365972847449118	0.4418737624580479
FP_25	FULL_PCC	dice_topk	40	0.3841702119706099	0.1739224612510571	0.3753987679435118	0.2585386907252648	0.49127557635214103	0.332306656083756	0.43858331440830334
MIXED	FULL_PCC	dice_topk	40	0.33275475996139303	0.1674463657291604	0.333740220675845	0.19975677037094075	0.45459091002382845	0.2829150068540406	0.3849300556428056
PARTIAL_25	FULL_PCC	dice_topk	40	0.32372225537501925	0.1641362693263026	0.3150063162784368	0.20184651062316797	0.43993323361161807	0.2765417290940611	0.3759876693129105
PARTIAL_50	FULL_PCC	dice_topk	40	0.3491819000755053	0.17049433522895496	0.3429842354867235	0.2192788219939305	0.4663558096759517	0.2976223797856957	0.40202622392143894
SHIFT_3	FULL_PCC	dice_topk	40	0.3684392010009529	0.17127513921471288	0.36299686423099475	0.24097492554005212	0.4824959646309791	0.3167095180859306	0.4213130199789713

Supplementary Table S8. Internal oracle-assisted target-volume-matched controls

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

method	metric	N	mean	SD	median
Fixed	topk_Dice	113	0.2682975790454364	0.1576598808729009	0.264132066063693
Fixed	topk_IoU	113	0.1648225486075349	0.11027699644301947	0.1521613833260249
Naive	topk_Dice	113	0.26859460336380714	0.15740052081181546	0.264132066063693
Naive	topk_IoU	113	0.16499282436654195	0.11013030778718455	0.1521613833260249
EIA-linear	topk_Dice	113	0.37014903519881887	0.22838276263711124	0.33870193591438585
EIA-linear	topk_IoU	113	0.2535643304510223	0.19291693300631707	0.20387788516568078
EIA-blend-0.90	topk_Dice	113	0.33882568432917093	0.17712419398426416	0.31979825614122936
EIA-blend-0.90	topk_IoU	113	0.21806651577704347	0.13444592460110105	0.1903332485784618
EIA-blend-0.75	topk_Dice	113	0.44381081129308003	0.20506428740777713	0.41276520353871715
EIA-blend-0.75	topk_IoU	113	0.3083613599283157	0.1797356337911695	0.26005302081037945
EIA-morph	topk_Dice	113	0.1907168793862762	0.12326047239491646	0.17770464846201056
EIA-morph	topk_IoU	113	0.11085339805060769	0.08102314081911022	0.09751693022629414
Full PCC	topk_Dice	113	0.43982123915672683	0.19370894098913222	0.4573386735338027
Full PCC	topk_IoU	113	0.3012818760976281	0.15906054213458906	0.2964608405569634
No-smoothing PCC	topk_Dice	113	0.6308292240200701	0.19583652400407495	0.6539280014217598
No-smoothing PCC	topk_IoU	113	0.48929591904861625	0.20463893963234114	0.4858046242080785

Supplementary Table S9. RHUH oracle-assisted target-volume-matched controls

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

method	metric	n	mean	SD	median	Q1	Q3	bootstrap_95ci_low	bootstrap_95ci_high
Fixed	topk_Dice	39	0.2066218552518812	0.13512876195298196	0.20843582697118956	0.09384659295903545	0.3038638860271052	0.16506380764110187	0.24836546661649436
Fixed	topk_IoU	39	0.12151667316552597	0.08607515528348274	0.11634293102142942	0.0492472680927187	0.1791636946299111	0.09516279586542364	0.14837109805320628
Naive	topk_Dice	39	0.2066218552518812	0.13512876195298196	0.20843582697118956	0.09384659295903545	0.3038638860271052	0.16506380764110187	0.24836546661649436
Naive	topk_IoU	39	0.12151667316552597	0.08607515528348274	0.11634293102142942	0.0492472680927187	0.1791636946299111	0.09516279586542364	0.14837109805320628
EIA-linear	topk_Dice	39	0.34345385436351666	0.2244980547413953	0.2917812183731811	0.1657196722098106	0.5270485267983904	0.2741324779418009	0.41343503564670175
EIA-linear	topk_IoU	39	0.23058901041394675	0.17778089314572484	0.17081021561260384	0.09040824830383903	0.35804724551773676	0.17639608511735233	0.2872711912715446
EIA-blend-0.90	topk_Dice	39	0.24075328158937945	0.14460822140971752	0.26190790760303473	0.10927267038065966	0.3453629770501446	0.19626563078741585	0.28512060530979766
EIA-blend-0.90	topk_IoU	39	0.1443946179239036	0.09448712154032146	0.15068701408340648	0.05782551460459358	0.20897322079322322	0.11544090381121841	0.17365236625996916
EIA-blend-0.75	topk_Dice	39	0.3459627291340022	0.2130440883300485	0.35300684509420804	0.18108121949937367	0.5317588006056149	0.2799466209013291	0.41210672523118047
EIA-blend-0.75	topk_IoU	39	0.22941841860224024	0.162709392815587	0.21433413007728647	0.09956302542500844	0.3621901788461197	0.17935356386295676	0.28063184014215786
Full PCC	topk_Dice	39	0.3612143407450775	0.17867368898946132	0.384335062084448	0.23817036721236723	0.4928786792537712	0.3055961685208271	0.4139500024614253
Full PCC	topk_IoU	39	0.23423203021277572	0.13090442249069195	0.23788042499892903	0.13518427426330748	0.32710171257181764	0.19398769649721406	0.2734054090240884
No-smoothing PCC	topk_Dice	39	0.4262530369110695	0.1966744793548497	0.42924693521139673	0.32686543155113734	0.5875575274633499	0.36506281261224244	0.48444043721307367
No-smoothing PCC	topk_IoU	39	0.28957554697135574	0.15526157327841086	0.2732746125705592	0.19555430004309227	0.41599049417526124	0.24211430702708495	0.3363838244023735

Supplementary Table S10. Failure and denominator accounting

Frozen descriptive evidence; no new scientific analysis was run for V2.1.

cohort	locked_denominator	analysed	scientific_failures
Development	40	40	0
Independent internal	113	113	0
RHUH external	39	39	0